Aagron School

MASTER DATASET REQUIREMENTS | OPERATIONS MANAGEMENT • COST INTELLIGENCE • EXPANSION STRATEGY

97 dataset categories across three school-management capabilities and their common AI foundation. This document lists the data required to develop, operate and evaluate the proposed systems.

1. Operations Management — 30 datasets

No.	Dataset name	Information required
1	School Master Data	School location, type, board, grades, sections, operating hours
2	Organizational Structure	Departments, reporting hierarchy, responsibilities, authorized decision-makers
3	Staff Master Data	Employees, qualifications, roles, subjects, departments
4	Staff Attendance	Present, absent, late, leave records, attendance patterns
5	Teacher Availability	Free periods, workload, substitutions, backup teachers
6	Class Timetable	Classes, subjects, teachers, rooms, periods
7	Academic Calendar	Holidays, examinations, events, academic deadlines
8	Student Attendance	Daily attendance, absenteeism patterns, class-wise records
9	Classroom Utilization	Total classrooms, capacity, occupancy, availability
10	Laboratory Operations	Laboratory schedules, equipment availability, utilization
11	Library Operations	Books, borrowing, returns, inventory, overdue records
12	Transportation Management	Vehicles, routes, schedules, delays, occupancy
13	Hostel Operations	Rooms, occupancy, maintenance, facilities
14	Asset Management	Equipment, furniture, computers, condition, location
15	Maintenance Management	Repair requests, breakdowns, maintenance schedules
16	Inventory Management	Stock levels, consumption, shortages, reorder points
17	Procurement Management	Purchase requests, orders, approvals, deliveries
18	Vendor Management	Vendors, contracts, performance, delivery reliability
19	Housekeeping Operations	Cleaning schedules, inspections, complaints
20	Security Operations	Security staffing, authorized access, incident records
21	Safety & Emergency	Fire safety, emergency procedures, incident response
22	Utilities Management	Electricity, water, internet, backup power, outages
23	IT Infrastructure	Computers, networks, software, downtime, support tickets
24	Events & Activities	School events, resources, schedules, responsibilities
25	Task Assignment	Task type, priority, assigned staff, deadlines

No.	Dataset name	Information required
26	Task Follow-up	Progress, reminders, delays, escalation
27	Historical Operational Problems	Previous incidents, causes, frequency, severity
28	Problem-Solution Outcomes	Suggested solutions, executed actions, success and failure
29	Scenario Simulation	Alternative solutions, constraints, estimated consequences
30	Operational Performance	Resolution times, recurring problems, service reliability, verified improvements

Most important training datasets: Historical Operational Problems, Problem-Solution Outcomes, Scenario Simulation, Task Assignment, and Task Follow-up.

These teach the Autonomous Brain how to identify a problem, investigate it, recommend an appropriate response, and track whether the authorized person has resolved it.

2. Cost Intelligence — 25 datasets

No.	Dataset name	Information required
1	School Financial History	Historical income, expenses, budgets, cash flows
2	Department-Wise Expenses	Academic, administration, transport, maintenance, IT expenditure
3	Budget vs Actual	Planned spending, actual spending, variances
4	Staff Salary Expenses	Salaries, allowances, benefits, overtime
5	Teacher Workload & Staffing Costs	Teaching hours, staffing requirements, substitute costs
6	Electricity Consumption	Monthly bills, meter readings, tariffs, peak usage
7	Water Consumption	Water bills, usage, leakage, maintenance costs
8	Internet & Communication Costs	Connections, plans, usage, recurring expenses
9	Transportation Expenses	Fuel, vehicle maintenance, insurance, driver expenses
10	Vehicle Utilization	Route-wise occupancy, distance, trips, unused capacity
11	Building & Facility Maintenance	Repairs, recurring service costs, maintenance contracts
12	Asset Purchase & Depreciation	Acquisition costs, useful life, depreciation, replacement
13	Equipment Utilization	Usage frequency, idle equipment, shared resources
14	Inventory Consumption	Quantity purchased, used, wasted, expired
15	Procurement Costs	Purchase prices, quantities, delivery costs
16	Vendor Pricing Comparison	Contract rates, alternative quotations, service quality
17	Software & IT Subscriptions	License costs, utilization, renewal dates
18	Printing & Stationery	Paper, printing, textbooks, stationery consumption
19	Laboratory Expenses	Chemicals, consumables, equipment, maintenance
20	Library Expenses	Book purchases, subscriptions, replacements
21	Hostel & Food Expenses	Meals, supplies, utilities, occupancy-related costs
22	Student-Wise Operating Cost	Total cost per student, cost per class, cost per section
23	Historical Savings Projects	Previous cost-saving actions, investment, actual savings
24	Cost Optimization Scenarios	Proposed changes, implementation costs, risks, expected savings
25	Savings Verification	Baseline spending, post-change spending, adjusted savings, quality impact

Most important training datasets: Department-Wise Expenses, Budget vs Actual, Historical Savings Projects, Cost Optimization Scenarios, and Savings Verification.

The AI should learn to identify why costs increase, calculate possible savings, compare options, and measure actual financial improvements.

For example, it may identify that several classrooms consume electricity while unused, estimate the possible savings from improved scheduling, and verify the reduction using subsequent electricity records.

Cost calculations, accounting totals, and budget limits should use deterministic software logic. The AI should focus on analysis and recommendations.

3. Expansion Strategy — 30 datasets

No.	Dataset name	Information required
1	School Profile	Location, board, school type, grades offered
2	Historical Student Enrollment	Year-wise, class-wise and section-wise enrollment
3	Current Student Strength	Current enrolled students by grade and section
4	Classroom Capacity	Available classrooms, seating capacity, occupancy
5	Total School Capacity	Maximum permitted and practical student capacity
6	Vacant Seat Analysis	Available seats by class, section and academic year
7	Historical Admissions	Applications, admissions, acceptance rates
8	Admission Enquiries	Enquiry dates, grades requested, enquiry sources
9	Admission Conversion	Enquiries converted into applications and enrollments
10	Waiting List Data	Students waiting, requested grades, demand
11	Student Retention	Continuing students, withdrawals, transfers
12	Historical Enrollment Growth	Year-wise growth and decline
13	Class-Wise Demand	High-demand grades, low-demand grades
14	Local Population & Demographics	School-age population, area-wise growth, demographic projections
15	Local Education Demand	Available education demand indicators, parent preferences
16	Competitor School Data	Nearby schools, curricula, fees, facilities, available public data
17	School Fee Structure	Grade-wise fees, annual fee revisions
18	Historical Fee Revenue	Tuition and other school revenue by year
19	Faculty Capacity	Teacher availability, staffing ratios, subject requirements
20	Infrastructure Availability	Land, classrooms, laboratories, toilets, transport, utilities
21	Expansion Regulations	Applicable recognition, affiliation, building and safety requirements
22	New Section Planning	Additional sections, required space, staffing, resources
23	New Grade Planning	Grade expansion requirements, curriculum, infrastructure
24	New Campus Location	Candidate locations, accessibility, catchment demand
25	Land & Building Costs	Purchase, lease, construction, renovation estimates

No.	Dataset name	Information required
26	Expansion Investment	Capital expenses, staffing, equipment, working capital
27	Revenue Forecasting	Enrollment scenarios, fees, projected collections
28	Operating Cost Forecasting	Salaries, utilities, maintenance, transport, administrative expenses
29	Profitability & Break-Even	Operating surplus, cash flow, break-even enrollment, payback
30	Historical Expansion Outcomes	Previous expansions, expected results, actual admissions, costs, returns

Most important training datasets: Historical Admissions, Enrollment Growth, Class-Wise Demand, Capacity, Historical Expansion Outcomes, and Expansion Investment.

These help the AI recommend whether a school should add sections, introduce new grades, expand facilities, or consider another campus.

The AI should distinguish between a possible opportunity and a financially justified expansion. Forecasts must include uncertainty, competing explanations, and alternative demand scenarios.

4. Common datasets needed for all three systems

No.	Dataset name	Information required
1	School Data Dictionary	Standard definitions for every field and measurement
2	System Integration Mapping	Connect ERP, finance, attendance, timetable and other systems
3	Historical Decision Records	What was decided, by whom, and why
4	Expert-Verified Training Examples	Correct examples reviewed by qualified staff
5	Failed Decisions & Negative Outcomes	Teach the AI which decisions did not work
6	School Policies & Permissions	Define what each role and AI workflow can do
7	Root-Cause & Reasoning Examples	Teach evidence-based investigation
8	Scenario & Simulation Tests	Evaluate alternatives and consequences
9	Action Execution Logs	Record authorized changes, assignments and outcomes
10	Quality & Safety Evaluation	Prevent harmful decisions and unauthorized actions
11	Data Privacy & Security	Consent, access control, retention, auditing
12	Continuous Improvement	Track accuracy, verified results and approved model updates

The three systems should also share these 12 datasets.

Implementation note

These categories describe needed information, not 97 separate model-training files. Use verified examples for training, live school records for operational decisions, and access-controlled datasets for sensitive student and staff information. Calculations, authorizations, budgets and safety controls should be enforced in software.